10 (a) The grey seal, *Halichoerus grypus*, is an aquatic mammal that lives in the North Atlantic Ocean. It feeds on fish, which it hunts at depths of up to 70 metres.

Fig. 10.1 shows a grey seal.

Fig. 10.1

Diving to hunt for fish has an effect on the respiration of the grey seal.

A study was carried out to measure the blood lactate concentration of a grey seal before, during and after a dive in deep water.

Fig. 10.2 shows the results of this study.

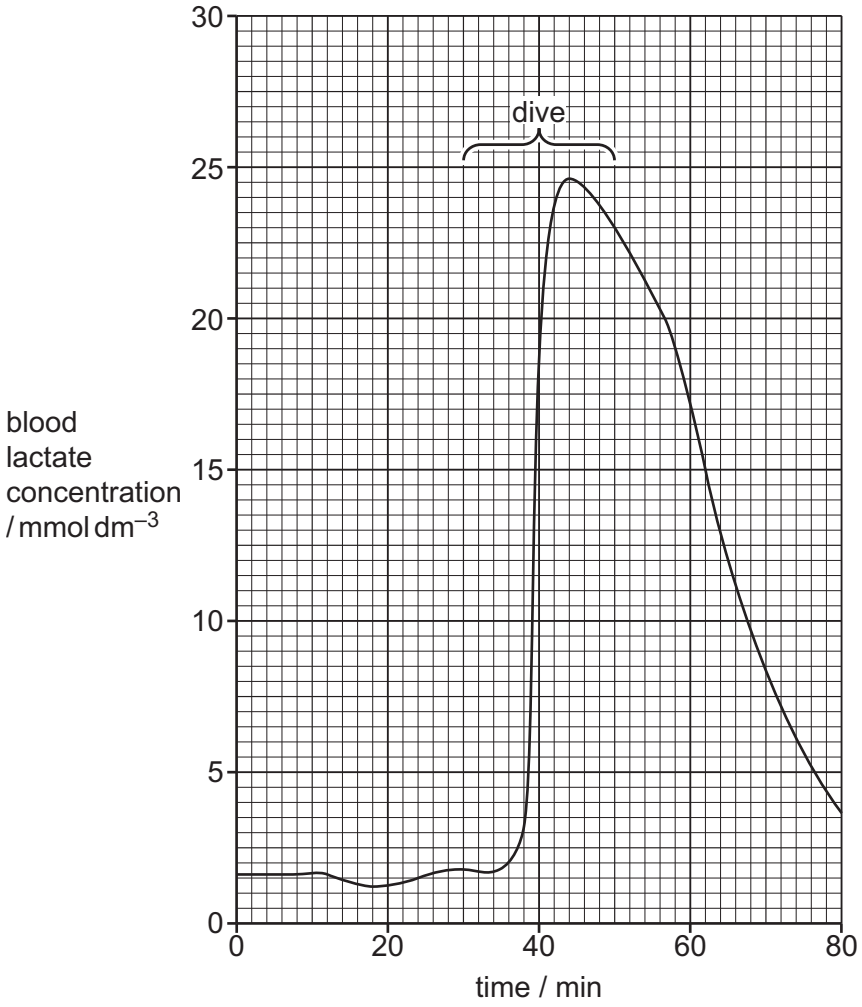

Fig. 10.2

With reference to Fig. 10.2, suggest reasons for the change in blood lactate concentration of the seal:

- during the dive
- after the dive.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Some seal species are classified as endangered on the IUCN Red List of Threatened Species™.

Suggest ways in which seal species may be conserved.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 7]

The boundaries and names shown, the designations used and the presentation of material on any maps contained in this question paper/insert do not imply official endorsement or acceptance by Cambridge Assessment International Education concerning the legal status of any country, territory, or area or any of its authorities, or of the delimitation of its frontiers or boundaries.

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.